

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2027

East Lansing, MI

- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Co-Founder & Vice President

Oct 2025 – Present

East Lansing, MI

- Eliminated manual onboarding entirely by automating member provisioning (GitHub API + Slack SDK) to grant full codebase and workspace access in 30 seconds, cutting setup time by 97% across 150+ members.
- Replaced paper sign-in sheets with a QR-driven check-in system backed by AWS S3 token caching, giving board members instant visibility into live participation data and cutting processing time by days.
- Crafted a cron-scheduled event pipeline that cross-checks check-in records with enrollment and posts robust, per-event Slack reports—bad records never break the batch thanks to soft-fail, idempotent functions.
- Enforced a 4-tier role hierarchy (prospect → member → board → e-board) at the database layer using Postgres row-level security, so every new API route inherits access rules automatically with zero per-route auth boilerplate.

WFS Consulting Group

Web Development Intern

May – Aug 2025

Plymouth, MI

- Built the firm's main customer-facing website in Vue, iterating on Figma designs with a non-technical stakeholder and delivering a mobile-responsive experience with validated contact forms.

PROJECTS

MindMosaic — Knowledge Graph Explorer | *Three.js, Spring Boot, Neo4j, PostgreSQL* Aug – Nov 2025

- Built an interactive 3D knowledge graph explorer where users visually traverse, search, and filter interconnected ideas.
- Designed dual-database persistence (PostgreSQL + Neo4j) to model relational data as a navigable graph; relationship-type indexing cut p95 query latency 48% (850ms → 440ms), making multi-hop traversals 3× faster than SQL self-joins.
- Integrated dual-provider OAuth 2.0 (Google + GitHub) with httpOnly JWT cookies, refresh token rotation, and user-scoped access control — each user traverses only their own subgraph.
- Authored CI/CD pipelines and Dockerfiles for the polyglot stack (Spring Boot + React + Neo4j), enforcing branch protection with automated test gates and 85%+ coverage across 35+ tests on every merge.

Dadei — Ambient AI Assistant | *React, Vite, Electron, FastAPI, pgvector, WebSockets, Redis* dadei.app

- Built an ambient assistant that listens across phones and laptops, runs speaker recognition and vector searching to build calendar entries, reminders and messages/emails from natural dialogue.
- Sharpens fuzzy who-when-what references through hybrid retrieval on Postgres/pgvector, ranking episodic rows, summaries, and recent turns by semantic, lexical, and recency signals before Gemini runs.
- Keeps phones and laptops on one network in lockstep without polling — clients subscribe over authenticated WebSockets, and the backend fan-outs fresh transcripts and proposed actions in parallel.
- Voice uploads ack immediately while transcription, diarization, and speaker matching run out of band; digest idempotency and per-network locks keep two laptops in one room from forking one timeline.
- Turns each conversation close into grounded memories and actions: Gemini runs summaries and tool calls anchored with transcript quotes, persisted against the same retrieved context so outputs track what was actually said.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, TensorFlow, NumPy*

Devpost

- Won SpartaHackX; Cooperative, gesture-controlled wizard duel where body movements cast spells, attacks, and abilities.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Cypher, Golang, C/C++, Bash

Frameworks React, Vite, Vue.js, Three.js, Spring Boot, FastAPI, Node.js, Tailwind CSS, Electron

Databases PostgreSQL, Neo4j, Redis, MySQL

Cloud & Infrastructure AWS (EC2, S3, CloudFront), Docker, GitHub Actions, CI/CD Pipelines